

Anjana Raman

December 6, 2025

DS 4002

Predicting if a Certain Branded Food is Healthy or Not

Understanding whether the foods we consume are healthy is essential for our overall well-being, yet packaged foods often make this difficult. Many branded products list dozens of ingredients and it can be confusing to determine which items should raise concern and which indicate nutritional value. In this case study, you will take on the role of a data scientist tasked with designing a simple scoring system that predicts whether a branded food is *healthy*, *intermediate*, or *unhealthy*. You will work with a dataset of real USDA branded food products, each containing a product name and an ingredient list. You will explore how different design choices such as identifying which ingredients count as “positive” or “negative,” deciding how strongly each ingredient should influence a score, and determining whether long ingredient lists should incur penalties shape the final classification of foods. Using the provided starter code and documentation, you will experiment with adjusting ingredient weights and grouping ingredients into categories. As you tinker with these definitions, you will see how your decisions change the scores assigned to each food and how closely your classifications match a baseline system. By engaging with this case study, you will gain insight into how data scientists translate qualitative judgments about food quality into quantitative rules that impact how people understand and evaluate the products they consume. Your final goal is to design, justify, and reflect on your own ingredient-based rubric.

Github link: <https://github.com/anjanaraman04/DS-4002-CS3>